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1

Computer Aided Learning of Chemistry for Grade 7 Student in Central Philippine University Junior High School

CHAPTER 1

INTRODUCTION

Overview of the Current Study

The rapid evolution of information and communication technology (ICT) has changed the landscape of societies and education. These changes in the educational landscape allow the learners to control their learning process and the ready access to a vast amount of information over that the teacher has no power or control (Lam & Lawrence, 2002). Hence, in the fast-changing world, technology such as a computer is appropriate to be at the leading edge of education because teenagers today have grown up with a different digital landscape than their teachers and are seemingly inspired and motivated by different technology. Moreover, educationalists believe that computers create an opportunity to improve the creativity of the students (Aqda, Hamidi & Rahimi, 2010).

Today's education needs to transform and take advantage of technological innovations available. In response, school administrators, officials, and educational institutions are seeking creative and effective approaches to technology in education. Hence, most of today's academic institutions have allocated millions of pesos to purchase computer equipment and other technological innovations to maximize the advantage of technology in the teaching and learning process. In like manner, a huge amount of the budget of the schools is allocated to spend a series of training and seminars for the faculty and teaching staff to be fully equipped with the use of computers in teaching. The application of technology in the teaching and learning process is known as computer-aided learning. Computer Aided Learning (CAL) brings various benefits as a teaching-learning tool. It can provide self-directed learning for learners, the exercise of different kinds of senses, and represent content in a variety of media. Likewise, CAL is one of the learning styles in which students can acquire knowledge through computer programs that can help monitor their learning and at the same can be used to evaluate themselves (Ozmen, 2004).

Educational technology is the effective use of technological tools in learning and incorporates numerous types of media that deliver. A computer is a general-purpose device that can be programmed to carry out a set of logical operations. Computers as technological tools have been

2

used to teach, manage, show and communicate information which makes them unique compared to the other learning devices (Georges, (2001)). Computer assisted learning (CAL) is a method, which uses a computer in the teaching and learning, thereby strengthening students' motivation and education in the process. In other words, use of CAL creates a learning environment where a computer programs, or its application is used to assist in teaching and learning of a particular subject. This is because the use of CAL gives opportunities to both students and teachers to learn at their speed and combine active learning with CAL. (Daintith, (2004)).

According to Daintith and Wright (2008), over the last two decades, the use of CAL has become more and more widespread and important especially in difficult subjects, more specifically in the sciences. Several reasons have been advanced by researchers for the use of CAL to support educational programs, chemistry education included. Among these is the capability to provide information in form of text, graph, audio, video, picture, animation and simulation which aids in strengthening student's mastery level. There is a form of one-to-one instruction and the opportunity for the students to proceed at their own pace, repeating parts of the exercise as they wish, none of which are available in a didactic classroom situation. In addition, there is the added variety and, perhaps, novelty in CAL, along with the potential to use vivid and animated graphics, enabling three dimensional aspects, and other features to be viewed more realistically (Sanger and Greenbowe, (2000)).

Many people consider chemistry as a challenging subject, which can discourage students from pursuing their studies in the field. Chemistry instruction involves a combination of lectures, textbook readings, laboratory experiments, and assignments. However, despite these conventional methods, educators often encounter challenges in effectively engaging students and promoting deep understanding of the subject matter.

Chemistry instruction may rely heavily on passive learning techniques, such as lectures and textbook readings, which may not fully capture students' attention or cater to diverse learning styles. This traditional approach can sometimes lead to rote memorization of facts and formulas rather than fostering critical thinking skills and conceptual understanding. The limited availability of resources and hands-on learning opportunities can hinder students' ability to apply theoretical concepts in real-world contexts. Laboratory experiments, while valuable, may be constrained by time, equipment, and safety considerations, limiting students' exposure to practical applications of Chemistry principles.

Desired State of Technology

3

Most respondents, the researchers found through data collection and survey administration, expressed an interest in a feature that would allow them to study and practice chemistry tasks safely from their homes. But it lacks resources on the internet. With these results, the researchers wanted to create a tool that can be used to teach students and do chemistry activities remotely.

Based on data gathered from instructors handling the subjects, there should be a tool to supplement their delivery of the subject matter in Chemistry. A Computer-Aided Learning in Chemistry that should be accessible on the web and mobile applications which can assist in teaching the chemistry subject. This application should include modules that present the topics in video, slides and pdf format, activities for certain lessons, and assessment features.

In addition, it should also include a module that creates an account for both user and teacher. A profiling module that collects the personal information of the user. This application will serve as a tool for teachers and students in learning the subject chemistry.

Statement of the Problem

This study aims to address the following problem:

1. Students have little attention and are easily distracted during lectures and less class participation.
2. Students sometimes have difficulty executing an activity due to the danger that they foresee to happen.
3. There are limited activities which allow active participation of students and help assess their learning.
4. Teachers cannot easily identify students who were not able to complete an activity if there is a delay in the recording of activities.

Objectives of the Study

General Objective

To develop a Web-Based Computer Aided Learning for the Chemistry Subject for Central Philippine University Junior High School Students.

Specific Objectives

Specifically, the study aims to design and create;

1. A module wherein students can view videos and slides related to the lecture topics;

4

2. An animated simulation module that will allow students to perform realistic laboratory experiments without real life risk;
3. An activity module that will challenge the students problem-solving and analytical skills in Chemistry that will increase participation and assess learning of students;
4. A progress monitoring modules of activities completed by students

Theoretical and Conceptual Framework of the Study

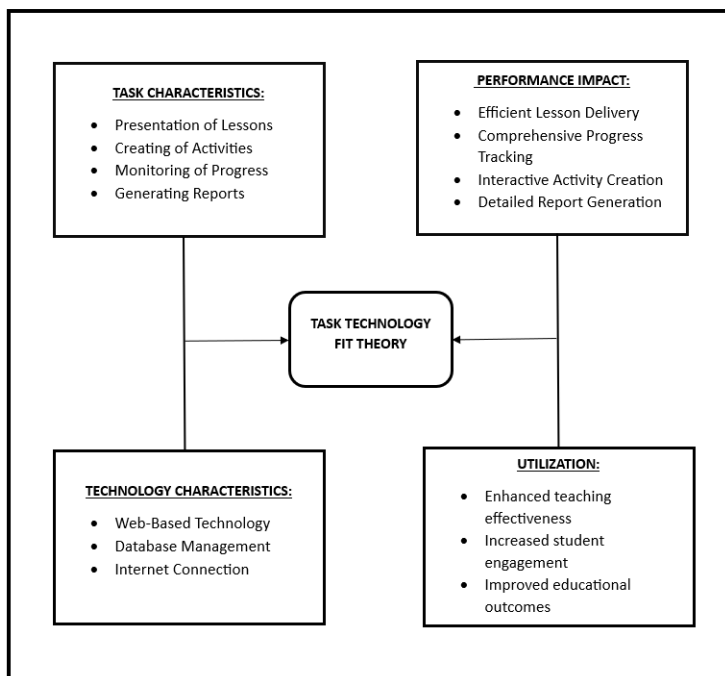


Figure 1. Theoretical Framework of the Study

The Task-Technology Fit Model was developed by (Goodhue & Thompson, 1995) to explain the utilization of technology by examining the fit of technology to users' tasks/requirements. The purpose of the theory was to add to the body of knowledge on technology utilization in the private and public contexts, which had limited explanation as to how the acceptance of technology contributes to

5

individuals' performance. TTF was the first theory that aimed to explore the post-adoption aspect of technology utilization, unlike other prior research, which had mainly focused on the antecedents of use and intention (Goodhue & Thompson, 1995).

Figure 1 Illustrates the Task Technology Fit Theory (TTF) the Task-Technology Fit Model, developed by Goodhue and Thompson in 1995, aims to explain how technology adoption is influenced by the alignment between technology and users' tasks and requirements. This theory seeks to expand the understanding of technology utilization in both private and public sectors, addressing a gap in previous research that focused primarily on adoption factors rather than on how technology contributes to individual performance. Unlike earlier studies that concentrated on the antecedents of technology adoption and usage intentions, the Task-Technology Fit Model was pioneering in exploring the outcomes of technology use post-adoption (Goodhue & Thompson, 1995).

The Task-Technology Fit theory posits that the performance of individuals or groups in completing tasks is significantly influenced by how well the technology employed supports the specific requirements of those tasks (Goodhue & Thompson, 1995). This theory underscores the importance of aligning the characteristics of tasks, such as complexity and interdependence, with the features of technology, including usability and functionality. A strong fit between task demands and technological capabilities leads to enhanced performance outcomes, such as increased productivity and user satisfaction. Task-Technology Fit theory is rooted in contingency theory, highlighting that the effectiveness of technology within organizations hinges on its contextual alignment with operational needs (newsletter.com).

The tasks and technological requirements of the Web-Based Computer-Aided Learning (Web-CAL) system for managing students' performance are outlined in Figure 1 shows how this theory was applied in the study.

The first model in Task-Technology Fit theory focuses on task characteristics, describing the tasks that users do to achieve their objectives. These processes usually involve presenting lessons, creating activities, monitoring progress, and generating reports.

In technology characteristics this pertains to the capabilities of the system itself. The objective is to enhance user experience by utilizing technologies such as web-based platforms to streamline and improve system efficiency.

The third is the performance impact, which encompasses the outcomes facilitated by the system, such as efficient lesson delivery, comprehensive progress tracking, interactive activity creation, and detailed report generation.

6

The fourth is utilization, which has positively impacted the system by enhancing teaching effectiveness, increasing student engagement, and improving educational outcomes. Utilization refers to how effectively the system is used in practice.

Conceptual Framework

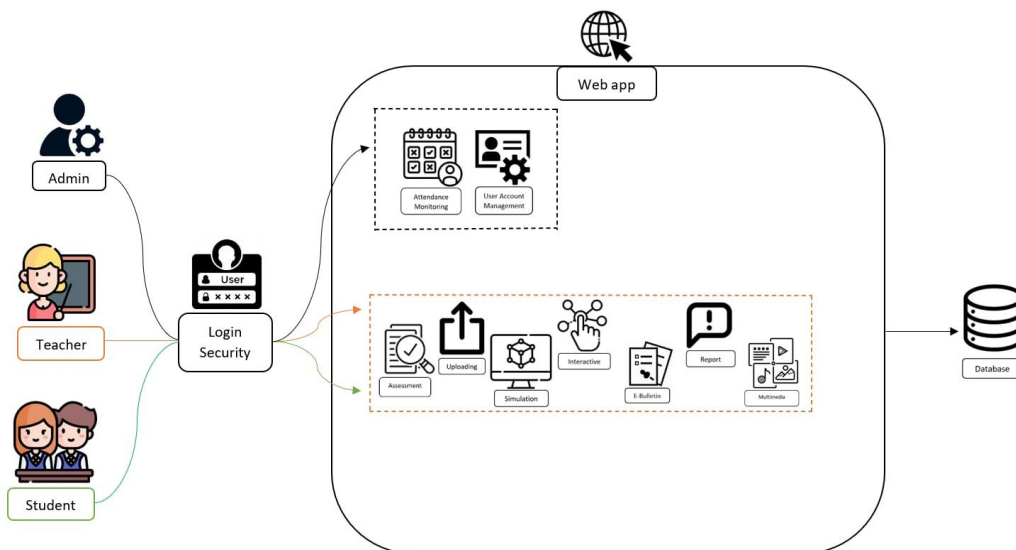


Figure 2. Conceptual Framework of the Study

Security

It refers to the measures and practices implemented to protect computer systems, networks, and data from unauthorized access, attacks, and damage. (www.techopedia.com)

This concept was applied to the process of log-in in which the system identifies the users to determine the access rights granted in the system. The administrator can access the

7

whole system. The teachers can view, and upload lessons and students can view lessons, submit assignments, and participate in discussions.

Web Application

A web application is a software application that runs on a web server, rather than being installed on the local computer. It is accessed through a web browser and typically designed to perform tasks over the Internet. Web applications can range from simple websites that provide information to complex systems with multiple functions and user interaction.

(www.dictionary.com)

This concept was considered as the core of how the whole system was designed and developed and the platform where the system users can access it. Such was applied to include the feature regarding the uploading and submission assignments even at the convenience of their home.

Account Management

The processes and procedures involved in managing user accounts within a system, network, or application. This includes activities like creating, modifying, and deleting accounts, as well as setting permissions and access levels. (techtargget.com)

This concept was used to ensure only authorized users can access the platform, they must verify their identity with a username and password. This helps keep the platform secure and protects everyone's information.

Attendance Monitoring

Attendance monitoring is the systematic process of tracking and recording the presence or absence of individuals, typically within an educational or organizational setting. It involves capturing data related to attendance patterns, duration of presence, and any deviations from expected attendance norms. The primary objective is to ensure accurate records for accountability, performance evaluation, and resource allocation purposes. (techtargget.com)

The concept was integrated into the system, which is managed by the Administrator, who oversees and verifies the attendance of all students enrolled in the courses. Chemistry teachers also have access to monitor and record student attendance within their respective classes. The system ensures accurate attendance tracking and accountability throughout the educational platform.

Multimedia

8

Integrates diverse media types like text, graphics, audio, and video in digital environments, enhancing user engagement through multiple sensory channels. It finds extensive use in education, entertainment, advertising, and communication to convey complex ideas, entertain, and create immersive digital experiences. Applications span interactive websites, educational software, virtual reality simulations, and multimedia art installations, facilitating interactive and impactful content delivery. (techtermsF)

This concept was applied in how the topics are to be delivered and presented. Such can enhance accessibility by providing alternative formats such as captions, transcripts, and audio descriptions. This ensures that all learners, including those with diverse needs, can access and benefit from the educational content.

Assessment

The process of evaluating or measuring the knowledge, skills, abilities, or performance of individuals, groups, or entities against predetermined criteria or standards. It aims to gather information about the extent of learning, achievement, or effectiveness of an educational program, training course, project, or organizational initiative. (www.merriam-webster.com)

This concept was applied and included in the system to check the students' comprehension in real-time. This helps educators tailor instruction based on individual learning needs and adjust teaching strategies accordingly.

Simulation

Includes developing a model or representation that resembles real-world processes, systems, or phenomena. This can be accomplished using a variety of methods, including physical models, mathematical models, and computer-based simulations. The goal of simulation is to provide a controlled environment in which scenarios can be investigated, experimented with, or explored in depth. It enables researchers, engineers, educators, and professionals to investigate complicated connections, estimate outcomes, and comprehend system behavior without having to physically engage with real-world situations, which can be costly, harmful, or impracticable to recreate. (techtargt.com)

This concept was implemented into the system to provide students with virtual laboratory experiences that replicate real-world experiments.

Interactive

9

Describe a system, application, or medium that actively engages users by allowing them to submit feedback and receive responses in real time. It promotes dynamic communication and engagement, allowing users to influence and change their experience based on their actions and decisions. Interactive systems frequently include interfaces or settings in which users may manipulate objects, travel through content, and engage in activities that respond to their inputs. This direct participation leads to a more immersive and engaging user experience, whether in educational software, multimedia applications, simulations, games, or digital learning environments. (www.merriam-webster.com)

Interactive refers to a process or system that allows for real-time interaction between users and a computer system. It involves two-way communication where users can actively participate and influence the system's responses based on their inputs. (techopedia.com)

This concept was considered part of the system's design to enhance student engagement but also supports active learning, critical thinking, and collaborative skills development essential in chemistry education.

E-bulletin

An e-bulletin is a digital communication tool used to disseminate concise, periodic updates, announcements, and information to a specified audience via electronic means. It typically integrates HTML or text-based formats for email distribution but can also be accessible through web platforms or mobile applications. E-bulletins are part of digital marketing and communication strategies, employing automated delivery systems and analytics to manage content dissemination, track engagement metrics, and ensure effective audience reach. (techtargget.com)

This concept feature serves to automate and streamline the process of delivering updates and announcements to users. It integrates with the system's backend to manage content creation, scheduling, and distribution, while providing analytics tools for evaluating the effectiveness of communications. This is crucial for maintaining user engagement and ensuring that users are informed of important updates.

Database

A database is a methodically organized collection of data typically stored electronically within a computer system. It is designed to efficiently handle large volumes of information by structuring data into tables, rows, and columns interconnected through defined relationships. Access and management of databases are facilitated by specialized software called a

10

database management system (DBMS), allowing users to securely store, retrieve, update, and manage data. Databases play a vital role across various sectors such as business, banking, healthcare, education, and government, providing a centralized and reliable means to store, access, and analyze essential information for decision-making and operational purposes. (techterms.com)

This concept provided a form on how content and data can be managed in the system. It allows for secure and efficient data handling through a Database Management System (DBMS), ensuring centralized storage and controlled access based on user roles. These practices support accurate information retrieval, analysis, and decision-making within the educational platform.

Scope and Limitations of the Study

The researchers' goal with this study is to equip both students and educators with effective technological tools and strategies to enhance the teaching and learning process.

The emphasis is on increasing the effectiveness of teaching Chemistry via digital means. Provide simple and easy to understand simulation tasks for Grade 7 students. The study covers the Chemistry subject and includes the following topics, Periodic Table of Elements, Ways of Separating Mixture, Introduction of Elements, and Classification of Mixture.

The application includes features such as uploading of lesson materials, creating of additional topics, create quizzes and activity, assessment, simulation of activity, feedback of quiz result, interactive activity related to periodic table of elements, and progress monitoring regarding completed.

The system does not include other topics covered in the subject outline of Chemistry. It cannot check submitted assignments or activities in essay form and activities created by teachers. The teacher must do the checking and grading manually.

Significance of the Study

The significance of developing a Web-Based Computer-Aided Learning (CAL) system for the Chemistry subject, specifically tailored for Grade 7 students at Central Philippine University Junior High School, extends beyond mere technological innovation. The study findings will therefore be beneficial to:

Chemistry Teachers. Provides teachers with effective methods for integrating Computer-Aided Learning (CAL) into their Chemistry lessons. By mastering the Web-Based CAL system, educators

11

can engage students more actively and improve learning outcomes, empowering them to modernize their teaching practices with technology.

Educational Researchers and Practitioners. Contributes to the body of knowledge surrounding technology-enhanced learning and instructional design. Researchers can gain insights into the effectiveness of Web-Based CAL systems in improving student engagement and achievement in Chemistry. Practitioners, including curriculum developers and educational policymakers, can use the findings to inform decisions related to curriculum development, educational technology integration, and instructional methodologies, ultimately enhancing the quality of education.

Central Philippine University Grade 7 Students. The study's primary beneficiaries, these students gain access to an innovative Web-Based Computer-Aided Learning (CAL) system tailored to enhance their Chemistry learning experience. By utilizing interactive multimedia resources, simulations, and customized exercises, students can deepen their understanding of complex Chemistry concepts, engage in self-directed learning, and improve their academic performance.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the relevant literature and studies that have significant bearings on the variables included in the research, as determined by an exhaustive and in-depth search conducted by the researchers. It focuses on several factors that will aid in the development of this study. The study focuses primarily on developing a web-based computer aided learning for chemistry. This study's literature consists of books, journals, articles, electronic materials such as PDFs, and foreign and domestic theses and dissertations that are deemed useful for the advancement of knowledge regarding the study.

Effectiveness of Computer-Assisted Learning in Chemistry Education: A Systematic Review and Meta-Analysis

Computer-assisted learning (CAL) has become increasingly prevalent in educational settings, offering various tools and platforms to enhance student engagement and learning outcomes. In chemistry education, CAL encompasses a range of digital resources such as simulations, virtual labs, multimedia tutorials, and interactive software designed to supplement or replace traditional classroom instruction. The study conducted by Smith and Brown (2020) aims to systematically review and analyze existing research on the effectiveness of CAL in chemistry education. By synthesizing data from multiple studies, the authors seek to provide insights into how different types of CAL interventions impact student learning outcomes, including conceptual understanding, problem-solving skills, and engagement with the subject matter. The study employed a systematic review and meta-analysis methodology. A systematic review involves a comprehensive search and critical appraisal of relevant literature to identify high-quality studies that meet predefined inclusion criteria. Subsequently, a meta-analysis statistically synthesizes

13

the results from selected studies to quantify the overall effectiveness of CAL interventions in chemistry education. (Smith, J. L., & Brown, M. T. (2020).

This systematic review and meta-analysis by Smith and Brown (2020) synthesizes empirical research on the effectiveness of computer-assisted learning (CAL) in chemistry education. The study examines a wide range of CAL interventions used in teaching chemistry, including simulations, virtual labs, multimedia tutorials, and interactive learning platforms. It analyzes the impact of CAL on student learning outcomes such as conceptual understanding, problem-solving skills, and engagement with the subject matter. The meta-analysis provides quantitative insights into the effectiveness of various CAL approaches and identifies factors that contribute to successful implementation in chemistry classrooms. This study illustrates how researchers evaluate the use of technology-enhanced learning tools in a specific subject area like chemistry, aiming to inform educators and policymakers about effective strategies for integrating CAL into science education.

Effect of Computer Assisted Learning on Students Long Term Development

The increasing penetration of computers and the internet has profoundly changed how firms and markets operate, and the education industry has been no exception. Technology also has the power to transform teaching by ushering in a new model of connected teaching. This model links teachers to their students and to professional content, resources, and systems to help them improve their own instruction and personalize learning. In addition to increasing productivity inside the classroom and updating the pedagogical paradigm, technology can improve the delivery of high-quality education to underserved areas. Specifically, computer-aided learning (CAL) can narrow the education gap between urban and rural areas, reducing inequalities in education achievement. (nber.org/system)

The Role of Computer-Aided Instruction in Science Courses and the Relevant Misconceptions of Pre-Service Teachers

Commented [GU1]: Aksakalli, A., Turgut, U. & Salar, R. (2016). The Role of Computer-Aided Instruction in Science Courses and the Relevant Misconceptions of Pre-Service Teachers. *Journal of Education and Practice*, 7(28), 67-71. from <https://www.learntechlib.org/p/195056/>.

14

In the study of Aksakalli et al (2016) examine how pre-service physics teachers interact with computers and identify any misconceptions they may have about computer-aided instruction. Recognizing computers as an indispensable tool in modern day education, the researchers designed computer-based physics course packages and prepared computer-aided assignments with the participation of the pre-service physics teachers. To assess the effectiveness of these computer-aided courses, Physics Achievement Tests were administered to the 50 pre-service physics teachers from the Physics Education Department of Kazim Karabekir Faculty of Education at Atatürk University during the 2015-2016 Fall semester. The study's findings, analyzed using SPSS software, revealed that computer-aided science courses were indeed effective in enhancing the educational outcomes of pre-service teachers. However, it also uncovered several misconceptions held by these teachers regarding computer-aided instruction. These misconceptions could potentially hinder the effective integration of technology in their future teaching practices.

In this research it underscores the importance of addressing and rectifying such misconceptions to maximize the benefits of computer-aided instruction in science education. By incorporating these insights into our own development of a Computer-Aided Learning (CAL) module for Grade 7 Chemistry at Central Philippine University, we can better prepare future educators. Ensuring that they have a clear and accurate understanding of the role and potential of technology in education will enhance the overall effectiveness and adoption of CAL modules in their teaching practices.

Knowledge construction in chemistry through web-based learning strategy

Smartphones and computer-based technology have revolutionized different development sectors including education. The current review describes the existing literature on Web-Based Learning (WBL) as a novel methodology applied in teaching and learning, and its potential effectiveness in teaching and learning chemistry. Further, the study intends to reflect

Commented [GU2]: <https://link.springer.com/article/10.1007/s10639-022-11369-x>

15

on how the WBL could be applied in the Rwandan context of teaching and learning chemistry as the country adopted the competency-based curriculum (CBC) in 2015. Web-Based Learning (WBL) encompasses interactive digital tools and resources that enhance educational experiences. Research indicates that WBL facilitates active learning, supports diverse learning styles, and improves student engagement and comprehension of complex chemistry concepts. Studies suggest that virtual simulations, online laboratories, multimedia tutorials, and collaborative platforms are particularly effective in fostering deeper understanding and practical application of chemistry principles. (Iyamuremye et al., 2022)

The article of Iyamuremye et al. (2022) emphasizes that knowledge construction in WBL is rooted in social constructivist principles. Learning is portrayed as a social and cultural process, where knowledge is constructed through interactions among learners and between learners and instructors. This aligns with Vygotsky's theory, which posits that cognitive development is influenced by social interactions and cultural tools. The review employs a narrative approach, synthesizing findings from various electronic databases. It evaluates the existing literature on WBL and its effectiveness in teaching chemistry, providing a broad understanding of its impact across different educational settings

The use of computer-aided learning in chemistry laboratory instruction

According to the study of Allred (2004) which investigates the creation and application of computer software for teaching chemistry labs at the University of Louisville in his dissertation. The research was focused on creating software that could instruct concepts and lab techniques without the need for traditional lecture formats. This method was designed to accommodate students who learn better through visual aids and to let students conduct experiments without having previously been exposed to the essential chemical concepts in their lecture courses.

In the said study it involved the development of three modules. The first module focused on using computer software to help introduce students to the concepts related to thin-layer

Commented [GU3]: Allred, B.R.T. The use of computer-aided learning in chemistry laboratory instruction. Ph.D. thesis, University of Louisville.
From: <https://www.learntechlib.org/p/125189/>

16

chromatography and setting up and running an experiment. This was achieved through digitized pictures, video clips, and written information. A review quiz was included to help reinforce the learned information. The second module was devoted to the concept of the “dry lab,” presenting students with relevant information regarding the chemical concepts and then showing them the outcome of mixing solutions. By these observations, they were to determine the composition of unknown solutions based on provided descriptions and comparison with their written observations. The third piece of the software designed was a computer game that followed the first two modules in providing information the students were to learn. The difference here was incorporating a game scenario for students to use to help reinforce the learning. Students were then assessed to see how much information they retained after playing the game. The findings in his indicated that technology-aided instruction significantly benefited student learning. However, the study emphasized the importance of employing multiple instructional methods to provide the best educational experience (Allred, 2004).

Allred's research provides valuable insights that can be directly applied to our thesis on developing a Computer-Aided Learning (CAL) module for Grade 7 Chemistry students at Central Philippine University. The benefits and lessons drawn from Allred's work can enhance our understanding of how to effectively engage students. Incorporating digitized media, such as pictures and videos, and interactive elements like quizzes and games, can make learning more engaging for Grade 7 students, who often respond well to visual stimuli. CAL modules allow students to learn at their own pace, which is particularly beneficial for young learners who may need more time to grasp complex concepts. Integrating quizzes and interactive exercises provides immediate feedback, reinforcing learning and helping students retain information better.

Molecular Design Studio: An Interactive Multimodal Chemistry Education Environment

The Molecular Design Studio is an innovative multimedia tool developed at Arizona State University to enhance chemistry education. This tool emphasizes user interaction and

Commented [GU4]: Erlandson, B., Dolgov, I. & Weatherspoon, E. (2007). Molecular Design Studio: An Interactive Multimodal Chemistry Education Environment. In C. Montgomerie & J. Seale (Eds.), Proceedings of ED-MEDIA 2007--World Conference on Educational Multimedia, Hypermedia & Telecommunications (pp. 3520-3525). Vancouver, Canada: Association for the Advancement of Computing in Education (AACE).
from <https://www.learntechlib.org/primary/p/25881/>.

17

audio-visual feedback by integrating interactive features within a hybrid physical-digital environment. A media database containing spoken information about organic compounds based on benzene rings is part of it.

The studio, created specifically for the ASU SMALLab, attempts to increase learning engagement and accessibility by simplifying complicated chemical concepts through visual and auditory aids. Thanks to interactive and multimedia components, it improves students' understanding of chemistry.

The application of the tool holds potential for raising student comprehension and engagement levels. Future plans call for incorporating it into other educational systems and broadening its application to include more chemical compounds.

The Molecular Design Studio could expand in more directions in the future, such as covering a wider range of chemical compounds with its application and incorporating it into other multimedia systems or using it as a stand-alone teaching tool. To evaluate the tool's influence on learning outcomes and investigate potential improvements to boost its instructional efficacy, an additional observational study will be conducted.

The creation of interactive multimedia tools that are similar to the Molecular Design Studio can be based on the information that was gathered during its development and implementation in different educational contexts. More specifically, by using these principles, web-based Computer-Aided Learning (CAL) modules for courses like chemistry can be created, providing students with a more interesting and productive learning environment.

Summary

CAL integrates technology into traditional teaching, offering personalized, self-paced learning, and immediate feedback. Studies show its effectiveness in improving student engagement and achievement. Additionally, CAL helps narrow educational gaps by delivering high-quality education to underserved areas, reducing inequalities in achievement. The

18

integration of CAL and web-based learning (WBL) in chemistry education significantly enhances student learning outcomes by improving conceptual understanding, problem-solving skills, and engagement. This assumption is based on the premise that CAL and WBL provide interactive, personalized, and resource-rich environments that support diverse learning styles and foster deeper comprehension of complex concepts. The Web-Based Computer Aided Learning in Chemistry for CPU Junior High School students aims to provide a personalized learning experience, enhancing student engagement and achievement through interactive and multimedia elements. In addition, it aims to promote long-term academic development, address educational disparities, and make high-quality chemistry education accessible to all students.

The studies where software was developed are outlined in Table 1 which highlight features and attributes suitable for advancing web-based Computer Aided Learning (CAL) systems in Chemistry. Each system articulates specific goals and objectives aimed at enriching educational processes for students of chemistry. The Web-based Computer Aided Learning System defines distinct functionalities tailored to help users achieve their educational objectives effectively. These capabilities encompass a diverse array of tools designed to aid users in their learning pursuits. Students can utilize a dedicated module to access videos and slides related to lecture topics, engage in realistic virtual laboratory experiments through an animated simulation module for safe practice, and participate in an activity module geared towards enhancing problem-solving and analytical skills in chemistry. In addition, a progress monitoring module tracks students' completion of tasks, providing a robust mechanism for comprehensively assessing educational progress. These integrated features significantly enhance the learning experience by offering comprehensive support to both students and educators throughout their educational endeavors.

The Molecular Design Studio uses user interaction and audio-visual feedback to simplify complex chemical concepts, improving student engagement and understanding. In order to make learning interesting for students in Grade 7, the researchers will incorporate a module into

19

the Web-Based Computer Aided Learning in Chemistry for CPU Junior High School students that consists of virtual lab experiments and interactive activities. By using visual and auditory aids to simplify complex concepts, it seeks to include interactive and multimedia elements akin to those found in the Molecular Design Studio, thus encouraging an effective and interesting space for learning.

According to the study of Allred (2004), using CAL to instruct students to learn chemistry laboratory instruction at the University of Louisville shows that multimedia tools in chemistry labs significantly enhance learning. Using a variety of instructional techniques is essential for success. Comparing the features in the application developed in the study of Allred, CAL for CPU has modules that provide additional features like user profiles and login security to guarantee a secure and personalized environment for learning. It also includes an assessment module that evaluates the learnings of the students. And in terms of content, the application developed covers a broad range of Grade 7 Chemistry topics and activities. These modules do not exist in the cited study. Virtual lab experiments and game-based learning are used in both the CAL for CPU and in the system developed for the study of Allred. These give students an interactive and interesting way to understand difficult concepts. Based on the cited study it only concentrates on laboratory concepts. Overall, despite similarities between the two systems, the CAL for CPU provides a more structured, safe, and all-encompassing learning environment.

Functionalities	CAL in Chemistry Lab Instruction (Allred)	Molecular Design Studio (MDS)	CAL in Chemistry for CPU
Personalized Learning	✓	✓	✓
Self-Paced Instruction	✓	✓	✓

20

Login Security			✓
User Profiles			✓
Multimedia Elements	✓	✓	✓
Virtual Lab Experiments	✓		✓
Assessment		✓	✓
User Interaction Focus	✓	✓	✓
Game-Based Learning	✓		✓
Database of Educational Content		✓	✓
Hybrid Physical-Digital Environment		✓	

21

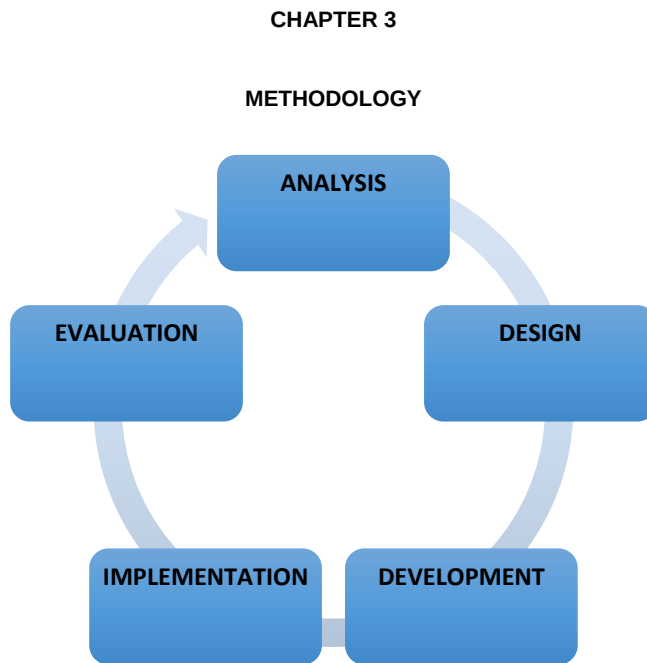


Figure 3. ADDIE Model Methodology

The ADDIE (Analysis, Design, Development, Implementation, and Evaluation) model is a fundamental framework in instructional design that offers a methodical way to create educational materials and memorable learning experiences. The model guarantees a well-organized development lifecycle through its five iterative phases, which makes it easier to create instructional materials that are effective.

In Figure 3, the researchers illustrate the application of the ADDIE model to their web application development by showcasing how each phase contributed to the overall process. The diagram elucidates the sequential nature of the ADDIE model, emphasizing completion before progression, yet allows for revisiting previous phases to validate and verify aspects of the project. This adaptability aligns with the iterative and evaluative nature of instructional design, making the ADDIE model a versatile and effective framework for creating educational solutions.

Analysis

The analysis phase involves analyzing the requirements gathered in the planning phase and identifying the specific features and functionalities that the web-based Computer Aided Learning content course module should have. This phase serves as the foundation of the methodology, involving an exhaustive exploration of requirements and objectives gathered during the planning stage. Utilizing a variety of research methodologies and interviews, the researchers meticulously assess learner needs and preferences, content requirements, and technological constraints.

To learn more about junior high school chemistry teachers' experiences, obstacles, and preferences regarding the use of educational technology in the classroom, The researchers conducted interviews with them. The interviews were designed with great care to ensure that the participants provided thorough answers. The researchers created a semi-structured interview guide with a variety of topics covered before the interviews, which included topics such as: their current teaching practices, their preferences for educational references and materials, perceived advantages and difficulties, and desired features and functionalities.

The teachers shared many viewpoints and experiences during the interviews. While some emphasized how technology can help create individualized learning experiences and engage students, others expressed concerns about technical difficulties and the need for continuous support and training. They highlighted that short-form videos were better to capture the attention of the students. These interviews yielded insightful information that influenced our analysis and later stages of the study. To ensure that the web-based CAL content course module meets the needs and expectations of educators and improves student learning outcomes, their opinions and feedback were extremely important in shaping its design and development.

Design

After the analysis phase, the CAL module's conceptualization and blueprinting are included in the design phase. Making use of knowledge gained from the analysis stage, the researchers identified that students preferred to engage with short-form videos, which led to the creation of a wide-array of videos covering specific topics in Grade 7 Chemistry along with interactive quizzes and supplementary learning materials like PowerPoint slides and word

23

documents to accommodate a range of learning preferences. The design also included a user-friendly dashboard inspired from other learning platforms such as Canvas, to ensure ease of access to topic materials and provide ease of navigation and familiarity to users. Based on feedback, the design also incorporated downloadable modules for students and teachers alike. As well as providing the ability for teachers to add, delete, or edit the provided learning modules to tailor it to their students' needs and ensure the material is not outdated.

The entire process was guided by the ADDIE model, as it allowed for continuous refinement and validation throughout the project's lifecycle. Different models were created to present the whole concept of the system. These include Use Case and Sequence diagram to show the different actions or processes that can be performed by users. Activity diagrams were also created to show the flow of each process integrated into the system. It was used to map out user interactions within the platform to identify user pathways and ensure a seamless experience. While crucial database schemas were designed through the entity relationship diagram to outline the data structure and relationships with the CAL platform to make sure that the platform could handle data efficiently and securely.

Development

This phase involves developing the web-based Computer Aided Learning content module based on the design created in the previous phase. The researchers started by combining the most important discoveries and figuring out recurrent themes and patterns. The features and functionalities that the CAL module should have to satisfy the needs and preferences of educators were determined using these insights as a basis.

The sequence of development began with the creation and structuring of the database. The database's purpose was to store user data, learning materials, and progress tracking information. By ensuring that the relationships and data structure were clear and effective, this important step gave the CAL module a strong backend.

After the database was developed, the menu and module creation sequence were carried out. The user authentication and access control system were the first modules to be created, guaranteeing safe login and suitable access levels for teachers, administrators, and students. This was followed by the user dashboard development, designed to be like commonly used learning platforms, to minimize confusion and provide easy access to course materials.

24

Next, the course content module is developed. One notable aspect highlighted during the analysis phase was the preference among teachers for short-form videos to capture students' attention and facilitate engagement. Building upon this insight, the researchers envisioned incorporating a comprehensive library of short, topic-specific video modules within the CAL platform. These videos were accompanied by interactive quizzes and supplementary materials to reinforce learning objectives and cater to diverse learning styles. The researchers also saw the necessity of consistent technical support and professional growth, so the researchers incorporated downloadable training modules and integrated support resources into the CAL platform. The researchers also designed the platform so that teachers and administrators have the power to edit, add, and delete learning modules to create their own digital library catered for their students. With the help of these resources, teachers would be able to make the most of the platform's features and increase the educational outcomes for their students.

The researchers made sure that the CAL module was reliable, user-friendly, and satisfied the needs and preferences of its users by adhering to this structured development sequence and including comprehensive testing at each stage. Testing was done at every stage of development. A module's functionality was tested through unit testing on individual components to verify that it worked correctly. Integration testing was performed to make sure that all modules worked together efficiently. While System testing verified that the CAL platform was functional and usable.

Implementation

This is the phase where the designed and developed educational materials are put into action. They are delivered to the target audience, be it students in a classroom setting, online learners, or any other user group. Making use of the knowledge gained from the analysis and design stages, The researchers now concentrate on introducing educators and students to the CAL platform, with the goal of fitting the CAL with their current learning system. To protect sensitive data and guarantee adherence to privacy laws, The researchers put strong user access controls and permissions management systems in place. According to their roles and responsibilities within the platform, educators, administrators, and students are all given the appropriate levels of access. To evaluate the efficacy and influence of the CAL platform on teaching and learning outcomes, ongoing monitoring and assessment are essential elements of

25

the implementation phase. The researchers improve user satisfaction and address identified needs by iterating on the CAL platform's features, functionalities, and design based on the results of monitoring and evaluation activities. Cycles of continuous improvement make sure the platform is responsive and adaptable to changing user needs and technology developments.

Evaluation

The last stage of the process is the evaluation phase, in which the researchers carefully evaluate how well the Web-Based Computer Aided Learning (CAL) content course module contributes to teaching and learning objectives. Building on the knowledge acquired during the phases of analysis, design, development, and implementation, the evaluation phase seeks to offer a thorough grasp of how well the module satisfies educator needs and enhances student learning outcomes. To assess the efficacy of the CAL platform in promoting knowledge acquisition, skill development, and academic achievement, the researchers carry out thorough assessments of student learning outcomes. Exams, quizzes, and pre- and post-assessments are used to track how well students have learned the material and have met their learning goals. Furthermore, in order to evaluate the platform's relative influence on learning outcomes, we compare performance data between students utilizing the CAL platform and those in conventional classroom settings. Surveying educators, administrators, and students about their experiences using the CAL platform to obtain qualitative information. These surveys evaluate user satisfaction, perceived usability, feature usefulness, and general satisfaction with the educational process. Open-ended questions give participants the opportunity to offer thorough criticism, recommendations for enhancements, and personal experiences of the impact they have on instruction and learning.

CHAPTER IV

WEB-BASED COMPUTER AIDED LEARNING OF CHEMISTRY FOR CENTRAL PHILIPPINE UNIVERSITY JUNIOR HIGH SCHOOL STUDENTS

System Overview

The Web-Based Computer Aided Learning in Chemistry for CPU Junior High School students' system is a tailored educational platform designed for Grade 7 students at Central Philippine University Junior High School. It caters to the diverse needs of teachers, students, and administrators, providing a comprehensive and engaging learning experience. This system represents a transformative approach to education, offering an interactive and adaptable platform aligned with modern pedagogical principles. It provides autonomy in assessment, accommodates higher-year students' review needs, and fosters a collaborative and engaging learning environment.

The system is designed to support meaningful learning in the context of chemistry education, utilizing web-based learning tools to enhance students' academic performance and facilitate the teaching and learning of chemistry. It aligns with the principles of computer-aided instructions and intelligent learning systems, providing a novel and effective approach to chemistry education. Additionally, the use of computer-aided learning systems in the field of chemistry has been a subject of academic research, demonstrating the potential for enhancing the learning experience and academic performance of students.

The system will store user information and utilize linear regression for processing their results. The system will provide suggestions for areas in which students need improvement. The administrative interface enables administrators to manage users and maintain the overall system.

System Objectives

The following are additional features which were included in the system.

1. A module that allows the teacher to upload video that can be used for discussion and simulated laboratory experiments virtually to be used for laboratory experiments.
2. A module that allows teachers to upload or create assignments and quizzes for students' assessment.
3. A module that allows students to view results of examinations to check their mistakes
4. A module that allows the students to download files that were uploaded by the teachers.
5. A module that allows teachers to see if students are opening other tabs that can lead to cheating.
6. A module that will provide personalized feedback to students based on their performance in quizzes and assessments, aiding in their learning and progress tracking.
7. A module that allows educators to track and assess students' performance in the assigned activities.

16

System Functions

Administrator

User Management. Create, modify, and delete user accounts for teachers, and students. Manage user roles and permissions within the system.

Content Management. Upload, update, and organize educational content such as lessons, multimedia resources, and assessments. Monitor and control access to content based on user roles and curriculum requirements.

System Configuration. Configure system settings, including security protocols, data backup procedures, and database maintenance. Customize the interface and layout of the system.

User Support. Provide support to teachers and students in case of technical issues or challenges related to system usage. Implement training programs for users to maximize the effectiveness of the system.

Instructors

Class Management. Create and manage virtual classrooms, including adding or removing students, assigning lessons, and monitoring progress.

Content Deliver. Deliver lessons using the system. Utilize multimedia resources, interactive activities, and other educational tools to enhance teaching.

Assessment and Feedback. Create quizzes, exams, and assessments to evaluate student understanding. Provide timely feedback to students on their performance and offer additional support as needed.

Progress Monitoring. Monitor the progress of individual students. Identify areas where students may be struggling and intervene accordingly.

17

Communication. Provide announcements, updates, and resources to enhance collaboration and engagement.

Students

Access to Learning Materials. Access and navigate learning materials, including lessons, multimedia resources, and reference materials.

Interactive Learning. Engage in interactive lessons and activities tailored to individual learning styles.

Assessment and Feedback. Take quizzes, exams, and assessments to gauge understanding. Receive immediate feedback on performance and track progress over time.

Progress Tracking. Monitor personal progress within the system. Access reports on completed activities, quiz scores, and overall performance.

Communication. Seek assistance or clarification on course content when needed.

Physical Environment

Hardware Specification

Table 2 contains the minimum hardware requirement for the server side.

Table 2. Hardware Requirement for Server Side

CPU Type/Model	Intel Core (™) i5(7th gen higher)
Input Devices	Mouse and Keyboard
Operating System	Windows 10 x64 bit and higher
Output Devices	Monitor and Printer

18

Storage Devices	4 TB (SSD, Solid State Drive)
Memory	8GB and higher

Table 3. contains the minimum hardware requirements for the client.

Table 3. Hardware Requirement Client Side

CPU Type/Model	Intel ® Core (™) i5-4440 CPU @3.10GHz
Input Devices	Mouse and Keyboard
Operating System	Windows 10 Pro x64 bit and higher
Output Devices	Monitor and Printer
Storage Devices	4.00 GB
Memory	4GB

Software Specification

Table 4. contains the software tools used in the development of the system.

Table 4. Software Specification

Operating System	Windows 10
Back End	MySQL, JavaScript and PHP
Front End	CSS, Node.js and HTML

19

Operating System: Microsoft Windows 10

Windows 10 is a Microsoft operating system for personal computers, tablets, embedded devices and internet of things devices. Windows 10 features built-in capabilities that allow corporate IT departments to use mobile device management (MDM) software to secure and control devices running the operating system. Windows 10's integrated search feature allows users to search all local locations, as well as the web simultaneously. The operating system also includes virtualization-based security tools. These Windows 10 features keep data, processes and user credentials isolated to limit the damage from attacks. (techtargert.com). This operating system was used in the development and implementation of the system because in terms of its features it is also very light in reference to its requirements.

Back-end: MySQL

MySQL is an open-source relational database management system. It is based on the structure query language (SQL), which is used for adding, removing, and modifying information in the database. Standard SQL commands, such as ADD, DROP, INSERT, and UPDATE can be used with MySQL. MySQL can be used for a variety of applications but is mostly found on Web servers. MySQL commands can be incorporated into the PHP code, allowing part or all of a Web page to be generated from database information. (techterms.com)

This was used and chosen by the researchers to implement the database structure needed by the system and due to the researcher's familiarity with the said software.

Back- End: JavaScript

JavaScript is a programming language commonly used in web development. It is a client-side scripting language, which means the source code is processed by the client's web browser rather than on the web server. This means JavaScript functions can run after a webpage has loaded without communicating with the server. JavaScript code can be inserted anywhere within the HTML of a webpage. (techterms.com).

In developing the web pages, the said language was utilized considering that this is the most common tool being used in web development.

Back End: PHP

20

PHP Stands for Hypertext Preprocessor, a recursive acronym. PHP is a scripting language web developers use to create dynamic websites. It is often installed by default on Apache web servers, alongside MySQL as part of a "LAMP" configuration. PHP scripts can range from simple one-line commands to complex functions. Some PHP-based websites generate nearly all webpage content dynamically using a series of PHP scripts. Developers can create custom object libraries and import them into various PHP pages, like a compiled language. (techterms.com).

The said language was used in the development of features pertaining to management of folders and files, content, and management of system in the administrator side.

Front- End: CSS

CSS is a style sheet language used for formatting content in HTML webpages. CSS style sheets can define the appearance and formatting of text, tables, and other elements separately from the content itself. Styles may be found within a webpage's HTML file or in a separate document referenced by multiple webpages. CSS helps web developers create a uniform look across an entire website (techterms.com).

To format the web page layout, this tool was used to implement the design of the page's content.

Front- End: Node.js

Node.js is an open-source JavaScript runtime environment. It runs JavaScript code outside of a web browser, allowing web developers to use JavaScript as the back end of their web applications. Node.js is also cross-platform and runs on Windows, Unix, Linux, and macOS. Node.js is popular among web developers for several reasons. First, it is efficient for applications that require many simultaneous connections and can handle many requests at once. It also has a large ecosystem of open-source packages and libraries available through Node Package Manager, which allows developers to easily add new features by integrating existing modules into their applications. Finally, the ability to use one programming language instead of two reduces the complexity of a project and provides an easier learning curve. (techterms.com)

This language was injected into the Java Script with the document object model to process data and display messages and handle user side processes.

21

Back- End: HTML

HTML is short for hypertext markup language. HTML creates electronic documents displayed on the World Wide Web. Each page contains several connections to other pages called hyperlinks. Every web page you see was written using one version of HTML. HTML code ensures the proper formatting of text and images for your Internet browser. Without HTML, a browser would not know how to display text as elements or load images or other elements. HTML also provides a basic structure of the page, upon which CSS is overlaid to change its appearance. One could think of HTML as the bones of a web page, and CSS as its skin. (hostinger.com)

HTML stands for HyperText Markup Language. It is a standard markup language for web page creation. It allows the creation and structure of sections, paragraphs, and links using HTML elements (the building blocks of a web page) such as tags and attributes. It is also worth noting that HTML is not considered a programming language as it can't create dynamic functionality, although it is now considered an official web standard. The World Wide Web Consortium (W3C) maintains and develops HTML specifications, along with providing regular updates. (hostinger.com)

To create templates and forms to collect data, this language was used. Users enter the data needed by the system through the interface that was created using this language.

Cost of Development

Table 5. contains the cost associated with the hardware requirement.

Table 5. Hardware Development Cost

Equipment	Amount
1 Monitor	P 4,000.00
1 System Unit	P 30,000.00
1 Keyboard	P 300.00
1 Mouse	P 200.00
TOTAL	P 34,500.00

Table 6. contains the estimated cost of the software developed.

Table 6. Software Development Cost

22

Researcher Role	Amount
Project Leader	P 13,000.00
Programmer	P 26,000.00
Researcher	P 13,000.00
Designer	P13,000.00
TOTAL	P 65,000.00

System Depreciation

Hardware Depreciation

Table 7. Hardware depreciation of the study over five years

Annual Depreciation = (Total Cost / 5) for each item.

Annual Depreciation Rate = (Annual Depreciation / Total Cost) * 100 for each item.

Year - The number of years since the start of depreciation.

Cumulative Depreciation = (Annual Depreciation * Year).

Table 7. Hardware Depreciation

Year	Amount
0	P 34,500.00
1	P 34,500.00
2	P 20,700.00
3	P 13,800.00
4	P 6,900.00
5	P 0.00

(Annual Depreciation / Total Cost) * 100 for each item.

Year - The number of years since the start of depreciation.

Cumulative Depreciation = (Annual Depreciation * Year).

Table 8. Software Depreciation

Year	Amount
------	--------

Software Depreciation

Table 8.

Software depreciation of the study over ten years

Annual Depreciation= 10/Total Cost of each item

Annual Depreciation Rate =

23

0	P 65,000.00
1	P 58,500.00
2	P 52,000.00
3	P 45,500.00
4	P 39,000.00
5	P 32,500.00
6	P 26,000.00
7	P 19,500.00
8	P 13,000.00
9	P 6,500.00
10	P 0.00

Tangible and Intangible Benefits

Tangible Benefits

- Reduced spending on textbooks and

printed works.

- Reducing paper usage and waste
- Teachers save time through automated grading and assessment tools to evaluate student performance
- Easy access to educational materials for both teachers and students

Intangible Benefits

- Improved communication and collaboration between teachers and students
- Lesson planning and delivery is more flexible
- Diverse learning styles using various multimedia resources
- Increased awareness and understanding of students' progress

24

Architectural Design

Use Case Diagram of the Proposed

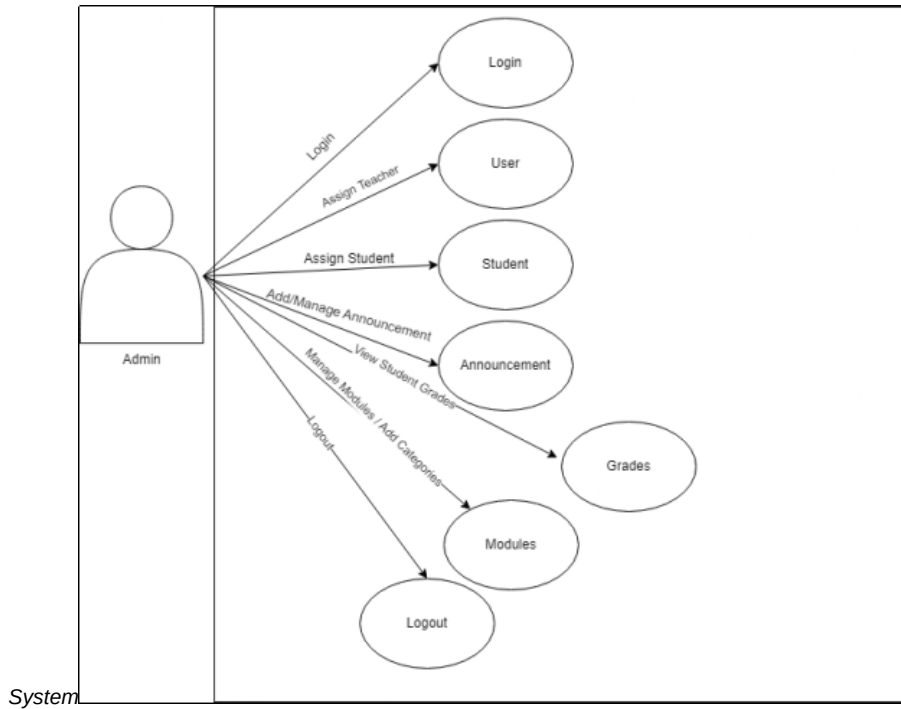


Figure 4. Use Case Diagram of the Proposed System -Administrator

Figure 4. shows the Use case diagram for the proposed for the Admin, Minister for organizations and Student organization president and vice-president. The system has 3 users, and each user has different contents accessible to them.

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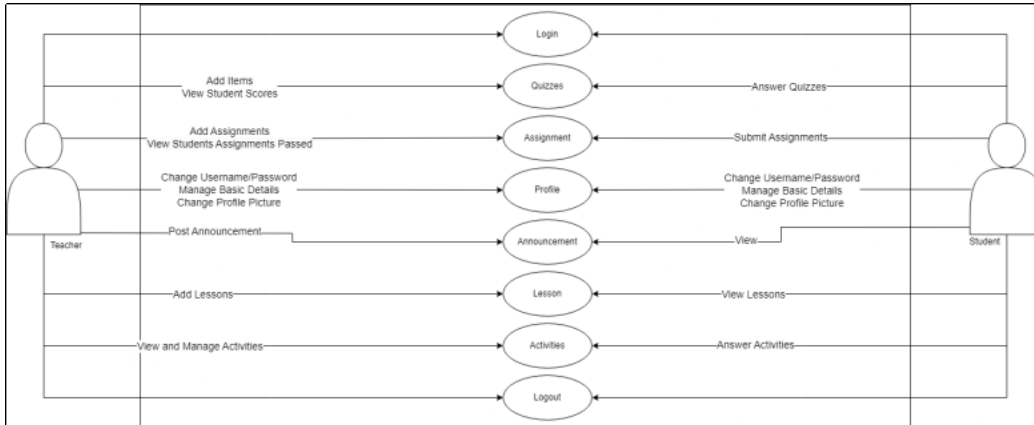
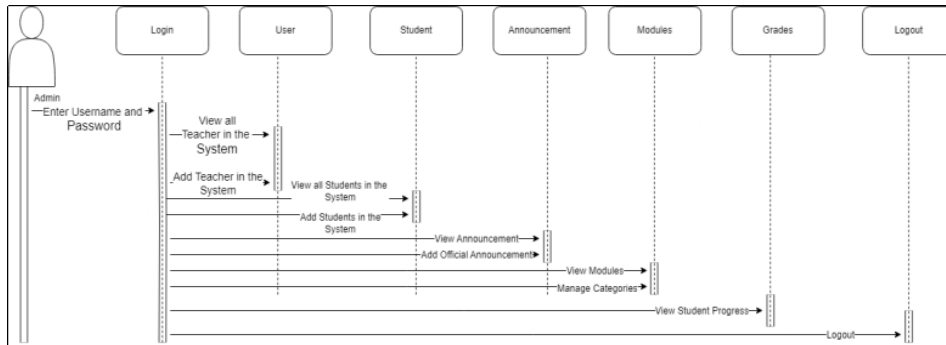


Figure 5. Use Case Diagram of the Proposed System- Minister for Organization and Student Organizations

Figure 5. shows that the Minister for Organization can change username and password. They can post both public and private announcements. In addition to posting activities, events, meetings, and awards, they can also view student organization applications and renewals. They can also monitor and confirm each student organization's attendance.

Sequence Diagram of the Proposed System

26



The Sequence diagram, Figure 6. shows the process and features for the Admin.

The administrator will access the platform by entering a valid username and password, which will then direct them to their personalized dashboard. In addition, the administrator can create and publish public announcements visible to all users. The administrator can also incorporate various educational components such as lessons, quizzes, and assignments, which can be utilized by the teaching staff. Moreover, the administrator will have the authority to modify the grading criteria used to assess student performance. This includes the ability to adjust the satisfaction grades required for evaluating students' progress.

The administrator role involves logging into the system using specific credentials, accessing a personalized dashboard, sharing public announcements, creating educational content, and adjusting grading criteria for student evaluation.

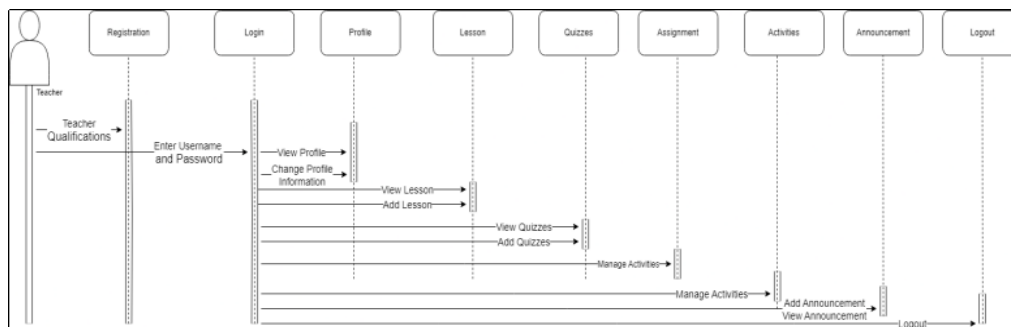


Figure 7. Sequence of Diagram of the Proposed System - Teacher

27

The sequence diagram, Figure 7, shows the process and features for the Teachers. Users assigned as teachers by the admin can login to the system using a unique username and password that will take them to the Teacher's dashboard. Teachers can upload multimedia files that serve as lessons or supplements to lessons, as well as assignments. Additionally, teachers can edit or delete existing lessons and assignments at their benefit. Teachers can also add new quiz questions to existing quizzes added by the admin or edit and delete existing ones.

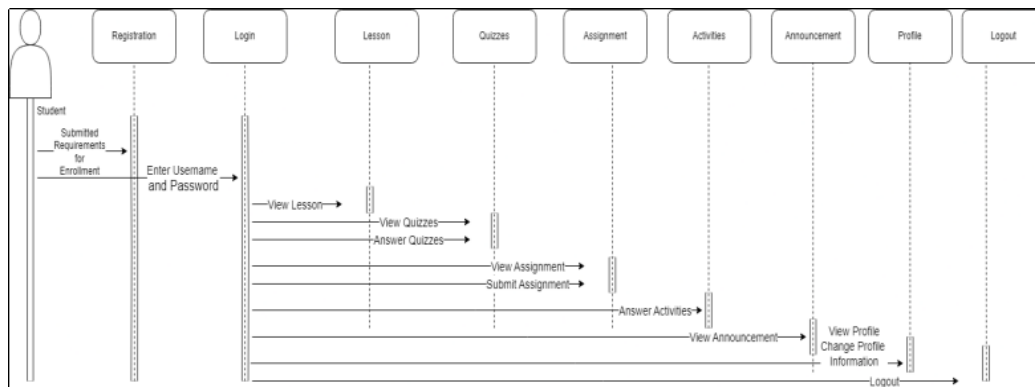


Figure 8. Sequence of Diagram of the Proposed System - Student

The sequence diagram, Figure 8, shows the process and features for the students. Users assigned as students by the admin can login to the system using a unique username and password that will take them to the student's dashboard. Students can access and download the multimedia files uploaded by Teachers that serve as lessons or supplements to lessons. Students can also upload their assignments as submissions for teachers to review. As well as view and take quizzes to evaluate their progress and interact with simulated activities.

28

Activity Diagram of the Proposed System

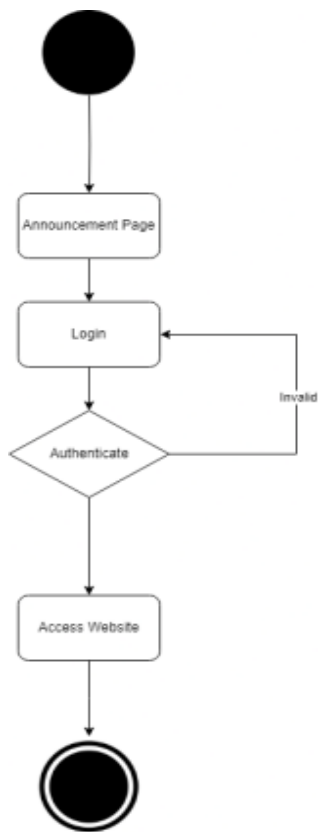


Figure 9. Activity Diagram - Access to System for Admin

The diagram illustrates how the users log in and use the system. To access the website and be routed to their dashboard, users must check in using their username and password if they do not have an account, they are unable to access the systems.

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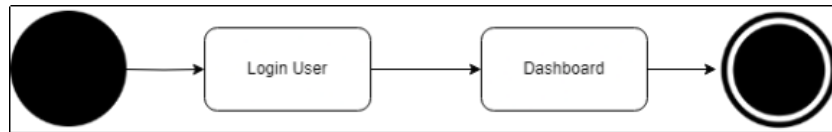


Figure 10. Activity Diagram - View Dashboard for all Users

Figure 10. shows the activity diagram for user log-in. Upon login the user will have access to the dashboard and the dashboard contains all the main menu.

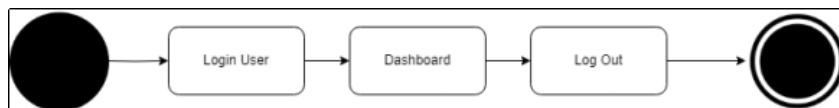


Figure 11. Activity Diagram - How to Logout Session for all users

Figure 11 shows the activity diagram for user logout. After Clicking the logout button, the user will direct to the home page.

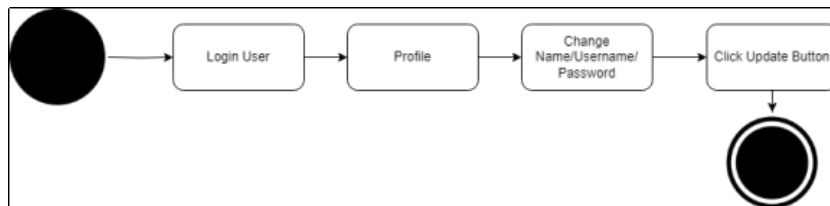


Figure 12. Activity Diagram - Change Name/Username/Password for All Users

Figure 5. shows the activity diagram for Change Name/Username/Password for all users. The user will navigate to Profile and Input all desired changes in Name, Username, or Password for the account.

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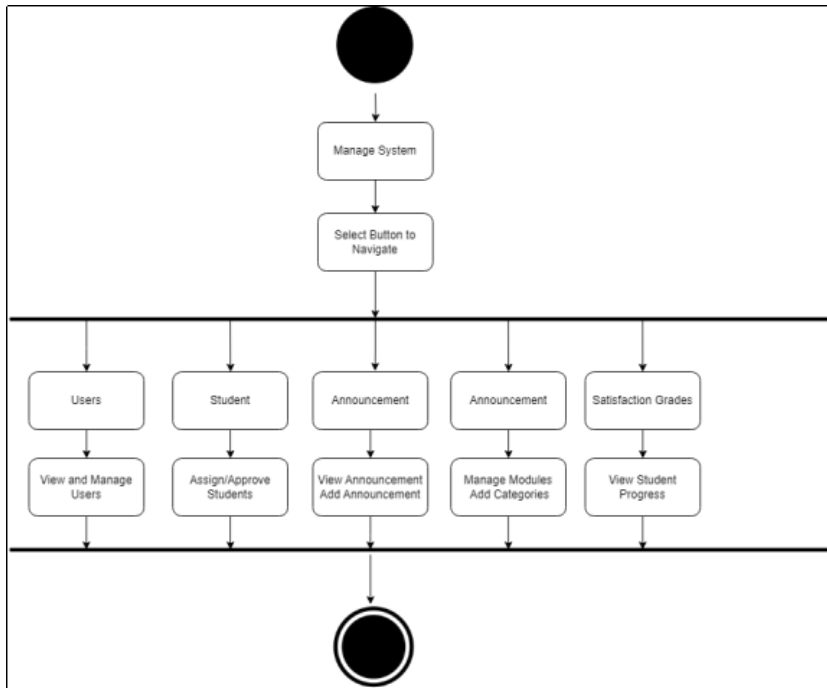


Figure 13. Activity Diagram of Admin for Manage Data

Figure 13. shows the activity diagram for Admin to manage all the Data of all Organizations.

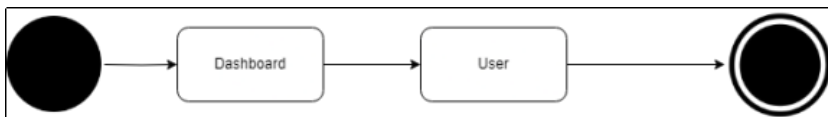


Figure 14. Activity Diagram of Admin to View all Users

Figure 14. shows the activity diagram for Admin to view all users in the system. The admin will click the user's button to view all the users in the system.

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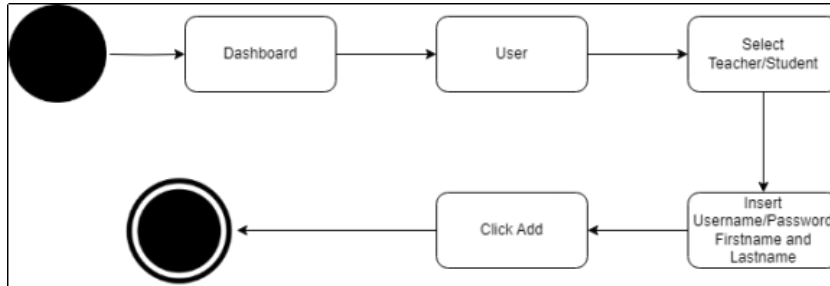


Figure 15. Add Students/Teacher

Figure 15. Shows how the admin can view all the users of the system and for them to be able to add them into the system by clicking their names, giving them username and password and finishing it by adding.

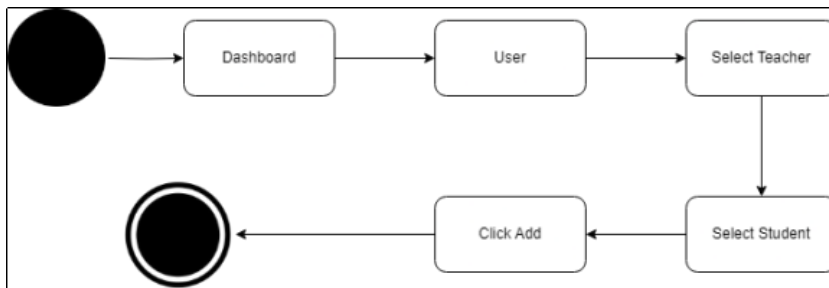


Figure 16. Activity Diagram - Post Announcement

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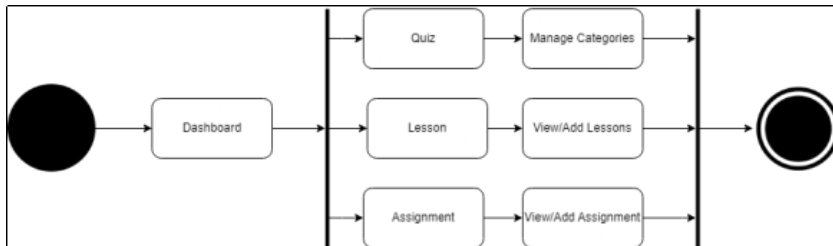


Figure 16 shows the activity diagram for both the admin and the teacher on how to post announcements. For Renewal the Administrator will decide whether the Renewal will be approved or not.

Figure 17. Activity Diagram of Admin to View all Approved Organizations

Figure 17. shows the activity diagram for Admin to View all Approved Organizations. The Administrator will click the Organizations button then navigate to the approved button.

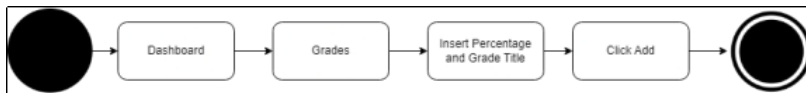


Figure 18. Activity Diagram - Modules Tab Admin and Teacher

Figure 18 shows the activity diagram for both Admin and Teacher. Both users can click on the modules tab and will choose which type of module they will navigate to. All options will give both teacher and admin the ability to add categories and quiz, view and add lessons, and lastly view, add assignments.

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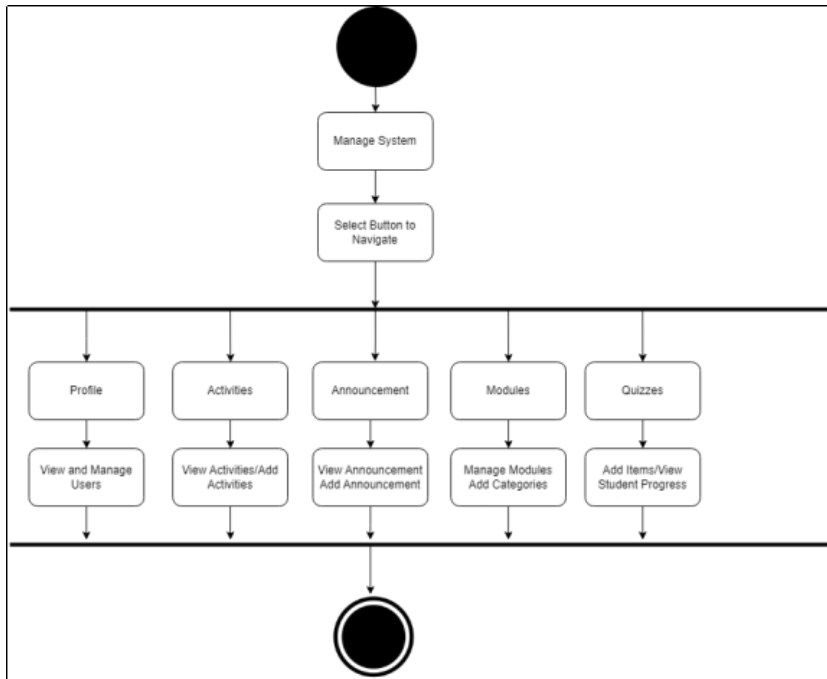


Figure 19. Activity Diagram of Teacher

Figure 19 shows the activity diagram for Admin to manage all the Data of Teacher.

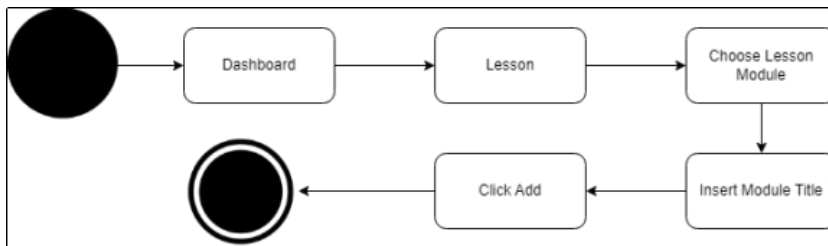


Figure 20. Activity Diagram – Manage Lesson Teacher

Figure 20 shows how the teacher can add lessons. They will have the choice to upload multimedia files for the students to view and download.

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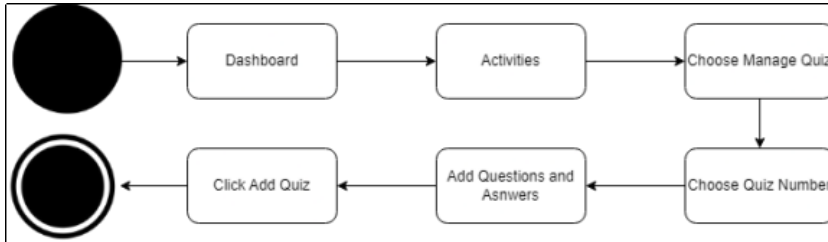


Figure 21. Activity Diagram - Manage Quiz Teacher

Figure 21 Activity Diagram shows how the teacher can add quizzes. They will choose first which lessons they will add the question and after adding the question, clicking add quiz will publish it and then it will add each question will have an option of timer, the timer of each question will be counted together to get the total duration for the quiz.

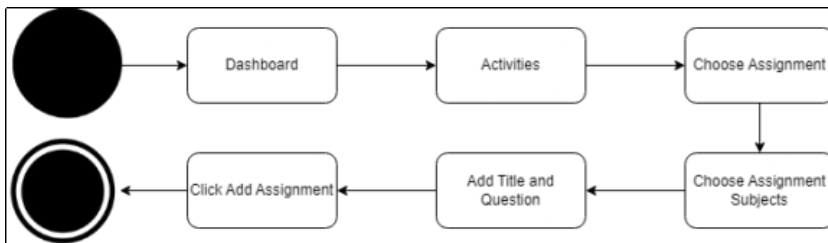


Figure 22. Activity Diagram - Manage Assignment Teacher

Figure 22 Activity Diagram shows how the teacher can add assignments. Choose the lesson on which the assignment will be set to, the teacher will have the option to set a due date or a deadline for the selected assignment.

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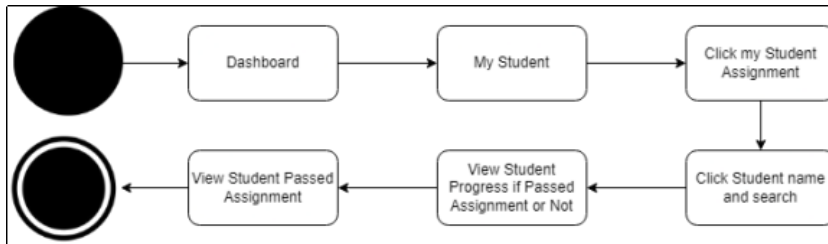


Figure 23. Activity Diagram - Manage My Students Assignments

Figure 23 Activity Diagram gives the teacher the ability to view assignments passed media files for assignment, and whether they have passed or not.

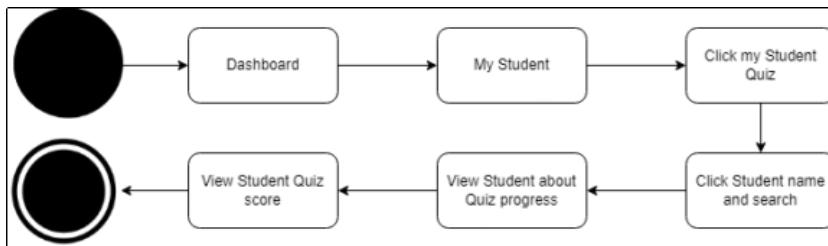


Figure 24. Activity Diagram - Manage My Student Quiz

Figure 24 Activity Diagram gives the teacher the ability to view quiz progress from students, whether they have taken the quizzes, passed, and their scores.

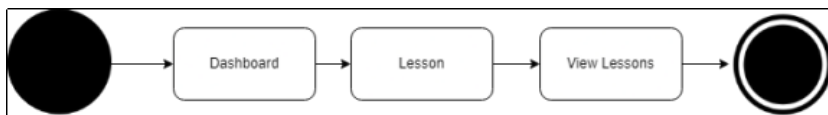


Figure 25. Activity Diagram - View Lessons Students

Figure 25 shows the activity diagram for the students to view lessons.

36

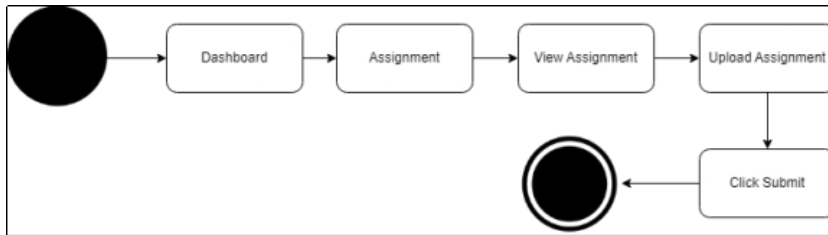


Figure 26. Activity Diagram - Assignment Students

Figure 26 shows the activity diagram for the students to view and submit assignments.

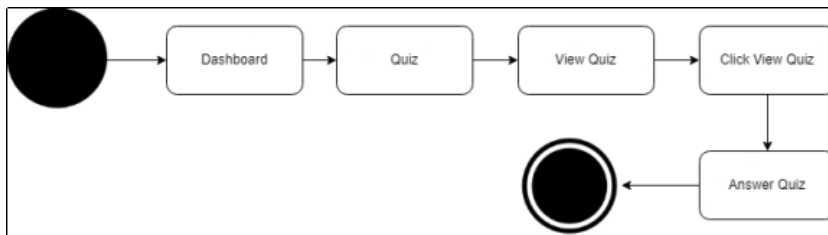


Figure 27. Figure 20 - Activity Diagram - View Quizzes Students

Figure 27 shows the activity diagram for the students to view and answer quizzes.



Figure 28. Activity Diagram - View Activities Students

Figure 28 shows the activity diagram for the students to view activities for certain lessons.

37

Database Structure

Database Name: calic
Table Name: user
Table No.: 1
Primary Key: id
Foreign Key: user_id
Table Description: This table stores the user data used to log in into the system.

Field Name	Data Type	Description
id	int	Login ID
type	varchar	Specifies if user is student, teacher or admin
user_id	varchar	User ID
username	varchar	Unique identifier for user
password	varchar	Alphanumeric phrase used to secure a user account
first_name	varchar	First name of user
last_name	varchar	Last name of user
photo	varchar	Image used as profile picture for user

Database Name: calic
Table Name: announcement
Table No.: 2
Primary Key: id
Foreign Key: user_id
Table Description: This table stores data used for public announcements in the system

Field Name	Data Type	Description
id	int	Announcement ID
title	varchar	Title of the announcement
user_id	varchar	User ID
description	varchar	Content of the announcement
date	varchar	Creation date of the announcement

Database Name: calic
Table Name: category
Table No.: 3
Primary Key: id

38

Foreign Key:

Table Description: This table stores data used for system modules

Field Name	Data Type	Description
id	int	Category ID
type	varchar	Type of module
name	varchar	User ID

Database Name: calic

Table Name: lesson

Table No.: 4

Primary Key: id

Foreign Key: user_id, category_id

Table Description: This table stores data used to create lessons

Field Name	Data Type	Description
id	int	Lesson ID
category_id	int	Category ID
user_id	int	User ID
title	varchar	Title of the lesson
files	text	Content of the lesson

Database Name: calic

Table Name: assignment

Table No.: 5

Primary Key: id

Foreign Key: user_id, category_id

Table Description: This table stores data used to create assignments

Field Name	Data Type	Description
id	int	Assignment ID
category_id	int	Category ID
user_id	int	User ID

39

title	varchar	Title of the assignment
files	text	Content of the assignment

Database Name: calic
 Table Name: student_assignment
 Table No.: 6
 Primary Key: id
 Foreign Key: student_id, assignment_id
 Table Description: This table stores data uploaded assignment files made by student users

Field Name	Data Type	Description
id	int	Student Assignment ID
student_id	int	StudentID
assignment_id	int	Assignment ID
files	varchar	Title of the assignment

Database Name: calic
 Table Name: quiz
 Table No.: 7
 Primary Key: id
 Foreign Key: user_id, category_id, lesson_id
 Table Description: This table stores data used to create quizzes

Field Name	Data Type	Description
id	int	Quiz ID
category_id	int	Category ID
user_id	int	User ID
lesson_id	int	Lesson ID
question	varchar	Quiz Question
choices	text	Quiz choices
correct_answer	varchar	Correct answer for the question

40

timer	int	Timer for quiz
files	text	Optional files uploaded to supplement the quiz question

Database Name: calic
 Table Name: student_quiz_session
 Table No.: 8
 Primary Key: id
 Foreign Key: user_id, category_id, lesson_id
 Table Description: This table stores data for students' quiz sessions

Field Name	Data Type	Description
id	int	Session ID
category_id	int	Category ID
user_id	int	User ID
lesson_id	int	Lesson ID
date_time_start	varchar	Date and time of when the session starts
date_time_end	text	Date and time of when the session ends

Database Name: calic
 Table Name: student_answers
 Table No.: 9
 Primary Key: id
 Foreign Key: student_id, quiz_id
 Table Description: This table stores data for quiz answers

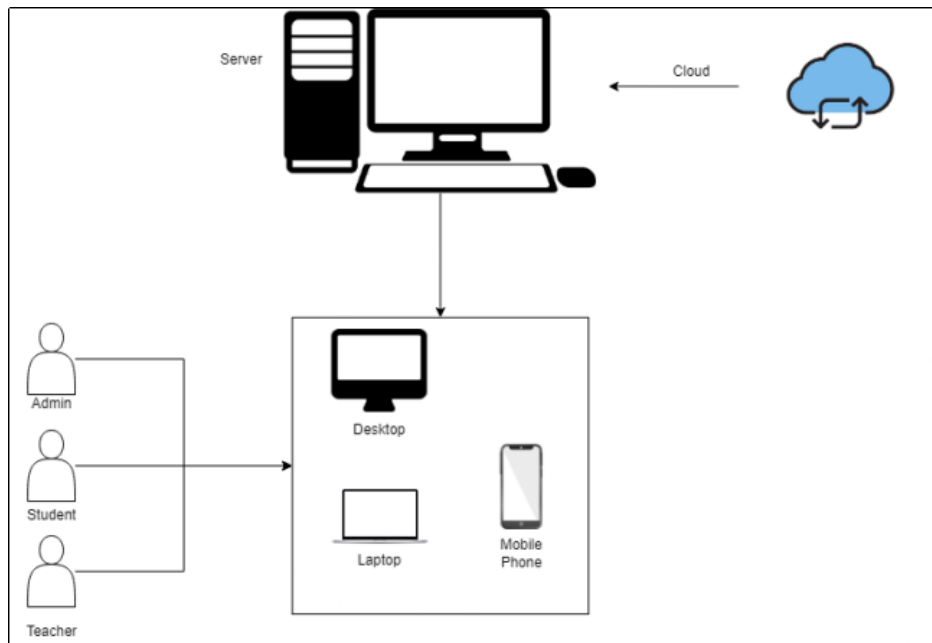
Field Name	Data Type	Description
id	int	Answer ID
student_id	int	Category ID
quiz_id	int	User ID

41

answer	varchar	Answer chosen by student
correct	int	This column determines if the chosen answer is correct

Database Name: calic
 Table Name: satisfaction_grade
 Table No.: 10
 Primary Key: id
 Foreign Key:
 Description: This table stores data used to set grades

Field Name	Data Type	Description
id	int	Grade ID
percentage	int	Percentage of grade
grade	varchar	Text description of grade



Network
Topology

42

Figure 29. Network Topology

Figure 29 is the system's network layout. The Web-Based Computer-Aided Learning System in Chemistry designed for Grade 7 students at Central Philippine University. The system caters to three distinct user roles: the administrator, teacher, and students.

Design and Implementation Issues

Web-Based Computer-Aided Learning System in Chemistry designed for Grade 7 students at Central Philippine University. It provides modules that cater to the diverse needs of teachers, students, and administrators, providing a comprehensive and engaging learning experience.

The following modules are for the administrator of the system. The first module manages new users by assigning them roles and providing them unique usernames and passwords for security purposes. Additionally, this module also allows the administrator to edit or delete existing users. The second module assigns a student user to a single teacher user, making virtual classrooms for ease of management and progress tracking. The third module is for making public announcements within the system, these can be seen by everyone enrolled in the system. These announcements can be purely text, or have external links attached to them. The fourth module allows the administrator to add new categories in Lessons, Quizzes and Assignments and edit and delete existing ones. These ensure all chemistry teachers are teaching similar lessons and let students have access to supplemental lessons provided by other teachers. The fifth module is for managing and setting the satisfaction grade on which students' progress will be gauged. Administrators can add new percentages and name them or delete existing ones to make room for new percentages.

The following modules are for users assigned as Teachers. The first module is to manage their profile by updating their username and password for security reasons and having the option to set their profile pictures. The second module lets teachers set up public announcements for their students, which can be text

43

or have external links attached. Teachers can also edit or delete their own announcement posts, but not the administrator's or other teachers. This module is like the administrator's announcement module but is limited. The third module allows the teacher to upload and manage multimedia files for lesson categories set by the administrator. All teachers can upload documents and multimedia files that students can view and download. These files can also be edited or deleted if needed via the edit or delete button. The fourth module is responsible for adding new quiz questions into the quiz bank. Teachers can add new questions to pre-existing quizzes set by the administrator by choosing their desired quiz on a dropdown menu. All teachers can edit or delete any question via the edit or delete button. The fifth module is where teachers upload new assignments for their students to work on. By choosing an assignment category set by the administrator, teachers can upload instructions in documents or multimedia files. These assignments can then be viewed by students on their dashboards. The fifth module is for teachers to manage and track their students' progress. Teachers can consult the assignment panel to see which students have submitted their assignments, and they can view or download these submissions via the view/download button. Teachers can also view which students have taken their quizzes and how well they have done in terms of scoring, the system will grade the student using the satisfaction grades provided.

The next modules are for users assigned as Students. The first module is to manage their profile by updating their username and password for formality and security reasons. The second module is for viewing lessons on the website via a pop-up window or having the choice to download documents or multimedia files for ease of access offline. The third module is for the viewing and submission of assignments or lab works. Students can select tabs related to their lessons and view instructions via the view button. Students can submit documents or multimedia files to their teachers via the submission button. The fourth module is the student quiz module, where students can view and take available quizzes to record their progress. Any quiz already taken will disappear from the list to avoid confusion. The last module is for interactive activities that students can access as supplementary material to their lessons.

Our system encountered several issues during development that required thoughtful solutions. One significant issue was the upload file size limitation. Users were unable to upload large files, disrupting the

44

workflow. This was addressed by implementing a condition that counts the bytes of each file. If a file exceeds the specified limit, the system denies the upload, ensuring that only files within the acceptable size range are processed. A login and reload issue also emerged, where users were redirected to the wrong page after logging in. This was particularly confusing and disruptive for users trying to navigate the system. To resolve the problem, this involved adding a controller for session management, which ensures that user sessions are correctly handled. Additionally, specifying the correct target page location ensured users are directed to the appropriate page upon login. The Table of Elements activity initially lacked interactivity, making it less engaging for users. To improve this, a value for each element was inserted, and a hover cursor function was added. These enhancements made the activity more interactive and user-friendly, allowing users to engage more deeply with the content. Lastly, there were challenges in developing a teacher-student chat system. Due to various constraints, a fully functional chat system could not be implemented. Instead, a one-way comment system was developed, allowing teachers to communicate with students regarding their assignments and quizzes. While not as dynamic as a chat system, this solution still provided a necessary communication channel between teachers and students, ensuring that feedback and instructions could be delivered effectively.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Summary

The focus of this study is to create a specialized Web-Based Computer Aided Learning (CAL) system tailored for chemistry education. Utilizing technology, it can be a tool that aids instructors on how to deliver and present content of Chemistry subject to Grade 7 students. Problems were stated in the study that pertains to students' attention during lecture and class participation, difficulty executing an activity, limited activities that engage active participation of students to assess their learning, and teachers' difficulty in monitoring students' progress.

The project's specific objectives include developing modules that incorporate video lectures and slides to enhance traditional lectures, as well as designing animated simulations that simulate realistic chemistry experiments in a safe environment. The study focuses on creating interactive activity modules aimed at improving students' problem-solving and analytical skills in chemistry. To ensure effective implementation and monitoring of these modules, the researchers have adopted a structured methodology, likely based on the ADDIE (Analysis, Design, Development, Implementation, Evaluation) model, emphasizing iterative development and evaluation to meet educational goals and enhance student learning outcomes.

The study employs the ADDIE model (Analysis, Design, Development, Implementation, Evaluation) for systematic development. Key methodologies include requirements analysis through interviews with teachers, ensuring the platform meets educational needs effectively. Design and development phases are informed by user preferences for short-form videos and interactive content.

46

The CAL platform integrates educational insights from teachers to offer a user-friendly interface resembling popular learning management systems. It includes modules for content delivery, interactive assessments, and teacher customization. Development focused on database structure, user authentication, and comprehensive testing to ensure functionality and security.

The study culminates in an implementation phase where the CAL platform is deployed, supported by rigorous evaluation methods to assess its impact on learning outcomes and user satisfaction. Continuous feedback and iterative improvements ensure the platform remains adaptive to evolving educational needs and technological advancements.

This comprehensive approach not only addresses current educational challenges but also aims to enhance student engagement and academic achievement in chemistry education through innovative technological solutions.

Traditional teaching approaches could not properly engage today's proficient in technology young people, thereby harming their chemical learning outcomes. There is a need for a modern, interactive learning platform that accommodates different learning styles and improves both student involvement and academic success.

Conclusion

The system accomplished the following objectives of the study as the researchers address the problems of the system's current state:

1. To create a module wherein students can view videos and slides related to lecture topics.

Chemistry teachers and researchers produced comprehensive video lectures and accompanying slides covering key chemistry concepts. These materials were integrated into the CAL system, providing students with accessible resources to reinforce their understanding at their own pace.

47

2. To develop an animated simulation module for realistic laboratory experiments. Innovative software for simulation was used to construct virtual laboratory conditions in which students could safely carry out experiments. This interactive component enabled hands-on learning experiences without the limitations or hazards of physical laboratories.

3. To design an activity module challenging problem-solving and analytical skills. Interactive activities were designed to present students with challenging chemistry challenges. These modules emphasized critical thinking and the application of academic information to real-life situations, encouraging active involvement and improving learning results.

4. To implement progress monitoring modules tracking students' activities. A comprehensive progress monitoring system was built into the CAL platform, allowing educators to track student involvement and measure learning progress in real time. This method is based on data enabled tailored feedback and support, resulting in ongoing improvement of student performance and curriculum delivery.

Recommendation

In developing the Web-Based Computer-Aided Learning in Chemistry for CPU Junior High School Students, it is important to consider the following recommendations for an improved information system in the future:

1. To innovate by incorporating more interactive elements such as virtual labs with even greater realism and interactivity. Consider including gamification features or collaborative tools to increase student engagement in the learning process.

2. To implement adaptive learning algorithms that personalize learning paths based on individual student performance and preferences. This approach ensures that students receive targeted support and challenges tailored to their specific needs.

48

3. To ensure efficient data management and security within the CAL platform, implement a scalable database architecture that can handle increasing volumes of student interactions, assessments, and content uploads.